VoltScope

Analog Voltage Deviation Visualizer

PROJECT OVERVIEW

Our team developed a circuit-based voltage deviation visualizer motivated by pressure-monitoring applications. A changing voltage emulates a pressure sensor's deviation from an optimal operating condition. The circuit rectifies and scales that signal, compares it with discrete thresholds, and displays the magnitude through coloured light-emitting diodes (LEDs). The visual response allows an observer to recognize low, intermediate, and high deviations without reading a numerical meter.

System Architecture

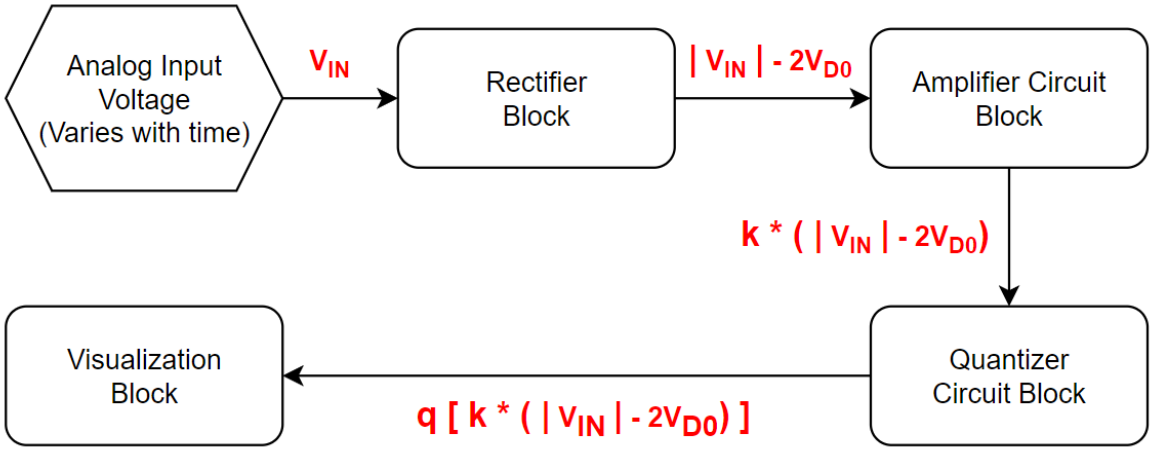


Figure 1: Original system block diagram from the CSE251 project handout.

Design and Operation

01 / Analog voltage input

The system accepts a **1 Hz** sinusoidal or triangular waveform. The amplitude and DC offset determine how the emulated sensor output varies around its optimum condition.

02 / Full-wave rectifier

A full-wave diode rectifier without a filtering capacitor converts positive and negative excursions into a unipolar deviation magnitude. The original block diagram includes the effect of diode forward-voltage drops.

03 / Amplifier stage

The rectified waveform is conditioned to remain within the required **0–5 V** operating range. The prescribed positive and negative supply/saturation bounds do not exceed **+8 V** and **−8 V**.

04 / Quantization and display

A resistor ladder provides reference thresholds for an op-amp comparator bank. The outputs control **eight LEDs**, indicating increasing voltage deviation through blue, green, orange, and red lights.

OUTPUT INTERPRETATION

Blue	Lowest voltage deviation from the reference condition.
Green / Orange	Intermediate deviation levels.
Red	Highest deviation; more LEDs are activated as the signal increases.

Conclusion

VoltScope illustrates the conversion of a continuously varying analog voltage into an intuitive, multilevel LED indication. The project integrates the fundamental electronics concepts of rectification, amplification, threshold comparison, and visual output in a pressure-monitoring-inspired application.

Original Circuit Schematics

The following diagrams are embedded directly from the Spring 2024 laboratory handout and show the reference circuits used for the voltage-level display.

Comparator-Based Quantizer

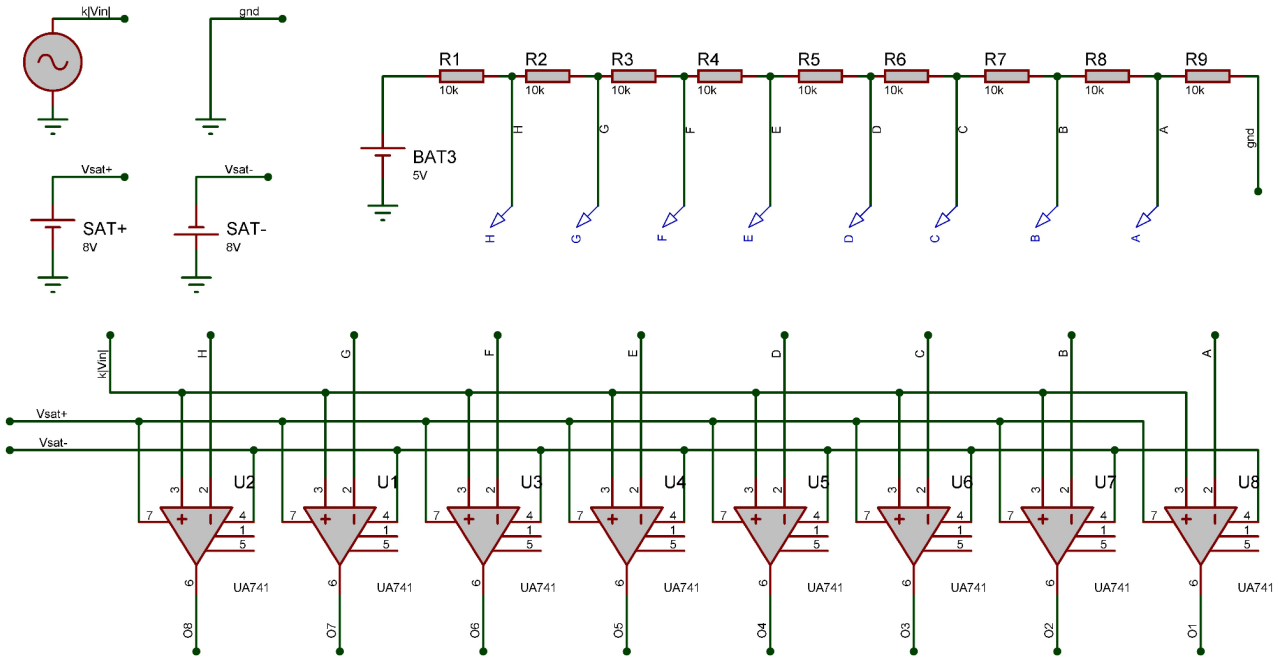


Figure 2: Original eight-comparator threshold circuit, including the resistor reference ladder.

LED Visualization Circuit

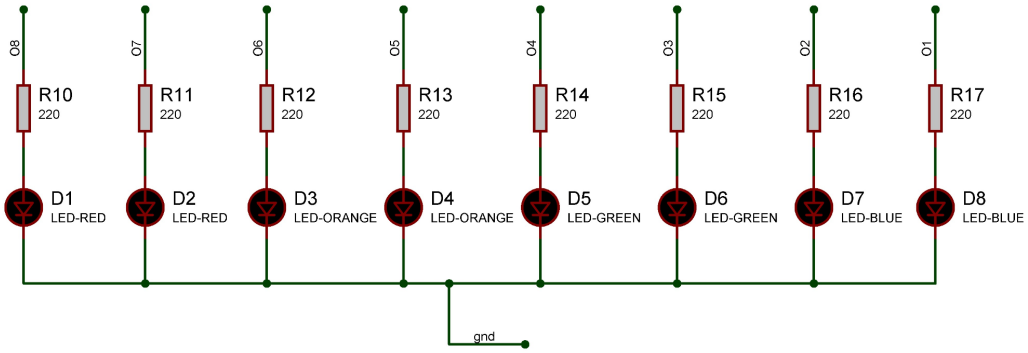


Figure 3: Original eight-LED indicator circuit with series current-limiting resistors.

Components and Checks

Element	Role
Diodes	Full-wave rectification
Op-amps	Scaling and voltage comparison
Resistors	Threshold generation and LED current limiting
Eight LEDs	Multilevel colour-coded indication
Breadboard/wires	Circuit interconnections

The design can be checked by applying a varying 1 Hz input, observing rectification and 0–5 V conditioning, and confirming that progressively more LEDs illuminate as the absolute deviation increases.

Op-Amp Reference

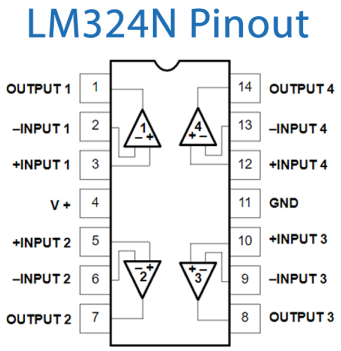


Figure 4: Original LM324 pin diagram.

The handout recommends the LM324 quad op-amp as an alternative to multiple UA741 devices.